

CSAI415 — Course Project Brief

PDF-Papers AI Agent: Hybrid Retrieval + GraphRAG with Online Learning and AutoML

Attribute	Details
Duration	10 weeks (4 h/week)
Project Weight	60% of module grade
Team Size	3–4 students
Tech (suggested)	Python, FastAPI, MongoDB, Qdrant (or similar vector DB), Neo4j, sentence-transformers, River, MLflow, Optuna/FLAML, PEFT/QLoRA, pytest, Docker Compose

Project objectives

Build an AI agent that answers questions over a corpus of scientific PDFs with grounded citations and page ranges, using hybrid retrieval (lexical+dense), a knowledge graph for reasoning (GraphRAG), lightweight online learning from feedback (River), and a small tuned model (PEFT/QLoRA).

Learning outcomes

- Design NoSQL schemas for AI workloads (document metadata, chunks, provenance, TTL).
- Implement hybrid retrieval and GraphRAG (graph-guided expansion + reranking).
- Use AutoML to optimize retrievers (e.g., kNN) or unsupervised cluster gating.
- Apply online learning with River to adapt to drift and user feedback.
- Fine-tune a small language model with PEFT/QLoRA and integrate it into the RAG pipeline.
- Evaluate faithfulness, relevance, latency, and show security mitigations (prompt-injection/retrieval-poisoning defenses).

Required system components

- Ingest: PDF → text → chunks (+overlap) + metadata (title, authors, venue, year, DOI, local path, page map).
- Stores:
 - MongoDB — chunk/doc metadata, provenance, run cards, evals.
 - Vector DB (Qdrant) — embeddings for chunks.
 - Neo4j — graph of Paper-(ABOUT)→Topic, Author-(WROTE)→Paper, Paper-(PUBLISHED_IN)→Venue (add CITES if time).
- Retrieval: BM25 or TF-IDF + dense (e.g., bge-small-en) → hybrid; cross-encoder reranking optional.
- GraphRAG: Use Cypher paths to find relevant subgraphs, expand to supporting chunks, blend with vector top-k, then answer with citations.
- Agent: Tool-using planner (e.g., simple ReAct/LangGraph style) with tools: vector_search, cypher_query, mongo_lookup, read_pdf_page_range.
- Adaptation: River online learner for query-to-topic classification or adaptive hybrid fusion weight; drift handled with ADWIN and lightweight triggers (no brokers).
- Tuning: PEFT/QLoRA for a 1–3B model on a small curated Q/A set from your corpus; quantize & cache for latency.
- Evaluation: Faithfulness, answer-relevance, Recall@k/NDCG@k, latency p95; safety (source pinning/provenance filtering).
- API/UI: FastAPI endpoints (/ask, /ingest, /feedback, /stats) + minimal UI showing answer steps and citations.

Dataset guidance

- 100–300 open-access PDFs from one topic (e.g., arXiv cs.AI/cs.CL subset, ACL Anthology subset, or departmental theses with permission).
- Provide a CSV with: paper_id, title, authors, venue, year, pdf_path/url, topics (optional).

Deliverables & weights

All deliverables should be through GitHub with commits from all the group members.

D1 — Week 5 (15%): Streaming Learner & AutoML Note

- AutoML Track (choose A or B):
- A) Supervised auto-tuned kNN retriever (Optuna/FLAML): search over k, metric, SVD dim, normalization, hybrid weight.
- B) Unsupervised clustering for cluster-gating (e.g., KMeans/spherical-kmeans): choose k, SVD, hybrid features; optimize internal metrics + a tiny labeled proxy if available.
- Online learning: River component for either (i) query→topic classification or (ii) adaptive hybrid weight from feedback (clicked helpful? y/n). Include drift handling (ADWIN) and prequential metrics plot.

- Short report (max 2 pages): Baseline vs AutoML metrics (NDCG@5/Recall@5), p95 latency; prequential chart; decisions and pitfalls.
- Repo: runnable notebook/script + YAML/JSON run card for the winning config.

D2 — Week 7 (15%): Retrieval Stack & Graph Build

- Ingestion pipeline: PDF→text→chunks→embeddings stored in Mongo + Qdrant; /search hybrid endpoint (BM25+Dense if possible).
- Neo4j graph: loaded nodes/edges with at least Authors, Papers, Topics; 3–5 example Cypher queries.
- Diagram: dataflow (ingest→stores→retrieval→graph).
- Metrics table: Recall@k/latency for /search on a small query set; top-k examples with citations.
- Repo: Docker Compose & seed scripts for stores; basic FastAPI.

D3 — Week 9 (15%): GraphRAG Executor, Evaluation & Safety

- GraphRAG executor: (1) choose subgraph by Cypher; (2) expand to supporting chunks; (3) hybrid blend and optional rerank; (4) answer with citations and page ranges.
- Evaluation: a small gold Q/A set, faithfulness & answer-relevance scores (RAGAS or equivalent), latency p95.
- Safety: implement at least one mitigation (e.g., source pinning, provenance filtering, deny risky tool calls); show before/after evidence.
- Ablation: vector-only vs graph-guided vs hybrid—compare quality/latency.

D4 — Week 10/11 (15%): SLM Tuning & Final Demo Package

- PEFT/QLoRA on a small (curated) Q/A set; include a tuning card (dataset size, epochs, lr, LoRA ranks, hardware/time, license).
- Integration: run zero-shot vs tuned inside GraphRAG; quantize & cache; final eval table (quality/latency deltas).
- Demo: 8-minute live demo + 8–10 page report (architecture, experiments, ablations, failure cases, ethics & licensing, future work).
- Repo hygiene: README with one-command setup, .env.example, seeds, test data; pytest smoke tests.

Grading rubrics

D1 — Week 5 (15%)

Criterion (Weight)	Excellent (A)	Good (B)	Satisfactory (C)	Weak (D/F)
AutoML design (6%)	Excellent (A): Clear search space; sensible	Good (B): Reasonable space; objective	Satisfactory (C): Narrow space or missing	Weak (D/F): Ad-hoc; no clear objective;

	objective (NDCG/Recall + latency); reproducible run card	mostly sound; minor reproducibility gaps	latency; partial reproducibility	not reproducible
Online learning (6%)	Excellent (A): River learner integrated; correct prequential metrics; ADWIN drift response justified with evidence	Good (B): Learner integrated; basic metrics; drift logic present	Satisfactory (C): Learner present but poorly evaluated	Weak (D/F): Missing or non- functional
Report quality (3%)	Excellent (A): Precise 2-page report; tables/plots; actionable insights	Good (B): Mostly clear; minor omissions	Satisfactory (C): Minimal narrative; few plots	Weak (D/F): Unclear; missing results

D2 — Week 7 (15%)

Criterion (Weight)	Excellent (A)	Good (B)	Satisfactory (C)	Weak (D/F)
Ingest & storage (3%)	Excellent (A): Clean chunking; robust PDF parsing; Mongo schema with provenance; Qdrant indexes	Good (B): Works; minor schema gaps	Satisfactory (C): Works but fragile parsing or minimal schema	Weak (D/F): Unreliable or incomplete
Hybrid retrieval (5%)	Excellent (A): BM25+dense with clear fusion; Recall/latency	Good (B): Hybrid works; some metrics	Satisfactory (C): Single modality or limited metrics	Weak (D/F): Poor retrieval; no metrics

	table; examples with citations			
Graph build (5%)	Excellent (A): Neo4j model sound; useful Cypher queries; data-store diagram clear	Good (B): Graph ok; 2–3 queries	Satisfactory (C): Minimal graph; trivial queries	Weak (D/F): Incomplete or incorrect
Engineering (2%)	Excellent (A): Docker Compose, FastAPI, scripts polished	Good (B): Minor rough edges	Satisfactory (C): Runs with effort	Weak (D/F): Hard to run

D3 — Week 9 (15%)

Criterion (Weight)	Excellent (A)	Good (B)	Satisfactory (C)	Weak (D/F)
GraphRAG pipeline (8%)	Excellent (A): Correct subgraph selection, expansion, hybrid blend, (re)ranking; consistent citations with pages	Good (B): Mostly correct; minor gaps	Satisfactory (C): Works but brittle or partial	Weak (D/F): Not working / missing
Evaluation (5%)	Excellent (A): Faithfulness & relevance computed; ablation thorough; latency p95 reported	Good (B): Metrics present; limited ablation	Satisfactory (C): Minimal metrics	Weak (D/F): No real evaluation
Safety (2%)	Excellent (A): At least one mitigation with	Good (B): Mitigation	Satisfactory (C): Rudimentary or ineffective	Weak (D/F): Missing

before/after
evidence; limits
documented

present; light
evidence

D4 — Week 10/11 (15%)

Criterion (Weight)	Excellent (A)	Good (B)	Satisfactory (C)	Weak (D/F)
Tuning & card (5%)	Excellent (A): PEFT/QLoRA well- configured; clear gains or justified parity; complete tuning card	Good (B): Works; modest gains; card mostly complete	Satisfactory (C): Marginal/unclear gains; sparse card	Weak (D/F): Broken or missing
Integration & perf (5%)	Excellent (A): Tuned model improves or maintains quality at lower cost; quantization/caching effective	Good (B): Integrated; some perf data	Satisfactory (C): Integrated with minimal eval	Weak (D/F): Not integrated
Demo & report (5%)	Excellent (A): Cohesive demo; 8–10 pp report with ablations, limitations, ethics	Good (B): Clear demo; solid report	Satisfactory (C): Adequate demo/report	Weak (D/F): Disorganized or missing elements

Global grading considerations (apply across deliverables)

- Reproducibility: seeds, environment, data paths; one-command run.
- Ethics/licensing: dataset licenses, privacy; citation of models/datasets.
- Teamwork: clear task ownership in README/report.
- Code quality: structure, typing/docstrings, tests.
- **AI Tool Engagement (applies to all deliverables)** This course actively encourages the use of AI tools throughout your project — for brainstorming, design decisions, debugging, evaluation, and writing. Thoughtful AI use is a professional skill you are expected to develop.
 - For every deliverable, each team member must submit a shareable link to their **complete, unedited AI chat history** relevant to that deliverable. Logs are not

penalised for wrong turns, bad prompts, or AI errors — in fact, these reveal the most about your learning process. Assessment focuses on:

- **Depth of engagement** — are you having substantive technical conversations, or asking for copy-paste answers?
- **Critical thinking** — do you push back, verify, iterate, and adapt AI outputs to your specific problem?
- **Individual contribution** — each member's link must reflect their own genuine involvement in the work.
- Submitting no logs, identical logs across members, or logs that do not correspond to the deliverable will be treated as missing evidence and will affect your grade accordingly.

Baseline targets (non-binding but recommended)

- Retrieval: Recall@5 ≥ 0.60 on your gold set; MRR $\uparrow$; p95 latency $\leq 2s$ on CPU-only small corpora.
- Answering: Faithfulness ≥ 0.8 , Answer-relevance ≥ 0.8 (RAGAS-style).
- Online learning: measurable improvement after drift or feedback ($> +5\%$ relative vs static on a temporal slice).
- Tuning: +3–5 pts in relevance/faithfulness or equal quality with lower tokens/latency.